Technical Report: Low-Level Memory Hardening in Constrained Environments

Author: Open Doors Applicant **Location:** Huambo, Angola **Domain:** Systems Security & Assembly Engineering

1. EXECUTIVE SUMMARY

> This report documents the practical constraints and engineering methodologies associated with identifying binary-level vulnerabilities and implementing low-level patch routines in embedded architectures. Conducted within a resource-constrained infrastructure in Huambo, Angola, the project focused on binary analysis, manual boundary checking in Assembly, and overcoming hardware isolation without dedicated vendor tooling.

2. PROBLEM STATEMENT & CONTEXT

In 2023, an investigation into legacy networking equipment and standalone surveillance controllers deployed across local infrastructure revealed recurring risks regarding unsafe memory management. Closed-source binaries operating on legacy ARM and MIPS chipsets frequently lacked modern compiler-level protections (such as ASLR or stack cookies), exposing systems to privilege escalation via memory corruption.

3. TECHNICAL EXECUTION & ASSEMBLY HARDENING

To address vulnerabilities without source code access, the development process required low-level binary analysis and surgical instruction patching:

- **Binary Disassembly:** Static analysis was conducted using open-source reverse engineering framework utilities to map stack frames and locate unverified buffer copying routines.
- **Assembly-Level Patching:** Hand-crafted Assembly routines were inserted to enforce strict boundary validation prior to instruction execution, overriding vulnerable function prologues directly in memory space.
- **Toolchain Adaptation:** Custom cross-compilation toolchains were built to target specific architecture variants (ARMv7-A / MIPS32) without relying on vendor-provided SDKs.

```
// Conceptual representation of boundary validation logic
CMP R1, R2 ; Compare input length against register limit
BHI truncation_label ; Branch to safe handling if input exceeds allocation
BL copy_routine ; Execute safe execution branch
```

4. REGIONAL ENGINEERING CONSTRAINTS

Developing low-level patches in Huambo presented specific logistical and operational barriers:

- **Hardware Isolation:** Lack of specialized debuggers (e.g., JTAG interfaces) forced reliance on remote GDB debugging over volatile serial links.
- **Infrastructure Stability:** Recurrent power supply interruptions disrupted continuous debugging sessions, requiring automated environment state backup scripts.
- **Tooling Scarcity:** Public datasheets for regional legacy chipsets were largely unavailable, necessitating manual register mapping.

5. CONCLUSION

This work demonstrated that security engineering at the Assembly level remains critical for legacy systems where source code is unavailable. Operating in resource-constrained environments reinforces the necessity of foundational system architecture knowledge, reliance on open-source toolchains, and disciplined low-level debugging skills.